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3.1.1. Numpy Array Operations

05:16

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size`

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `rows` and `columns`, representing the number of rows and the number of columns.
- The next `rows` lines contain `columns` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

```
import numpy as np

# Read rows and columns
r, c = map(int, input().split())

elements = []

# Read elements row by row
for _ in range(r):
    elements.extend(list(map(int, input().split())))

# Create NumPy array and reshape
arr = np.array(elements).reshape(r, c)

# Output
print(arr)
print(arr.ndim)
print(arr.shape)
```

```
print(arr.size)
```

Average time
0.014 s
14.42 ms

Maximum time
0.043 s
43.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 10 out of 10 hidden test case(s) passed

✓ Test case 1 43 ms Debug

Expected output

Actual output

3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[·1··2··3··4]]	[[·1··2··3··4]]
·[·5··6··7··8]	·[·5··6··7··8]
·[·9·10·11·12]]	·[·9·10·11·12]]
2	2
(3, ·4)	(3, ·4)
12	12

✓ Test case 2 6 ms